

Visible Buildings Queries

Time limit: 1.00 s

Memory limit: 512 MB

There are n buildings in a row numbered $1, 2, \dots, n$ from left to right. You are standing to the left of the first building. You can see a building if it is taller than all buildings to its left.

Your task is to process q queries: If only buildings in range $[a, b]$ existed, how many buildings would you see?

Input

The first line has two integers n and q : the number of buildings and queries.

The second line has n integers $h_1, h_2, \dots, h_n$: the heights of the buildings.

Finally, there are q lines describing the queries. Each line has two integers a and b .

Output

For each query, print one integer: the number of visible buildings.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq q \leq 2 \cdot 10^5$
- $1 \leq h_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
5 3 4 1 2 2 3 1 5 2 5 3 4	1 3 1